

```

articles = [
    """Apple unveiled its newest iPhone at an event in Cupertino on
    Tuesday.
    CEO Tim Cook said the device features a faster chip and improved
    camera.
    The phone will be available in stores starting next month, the company
    said, with prices starting at $999."""",
    """The World Health Organization announced new guidelines on Monday
    aimed at reducing sugar consumption worldwide. Officials in Geneva
    said
    the changes could prevent millions of cases of diabetes over the next
    decade if adopted by national governments."""",
    """Tesla reported record deliveries for the third quarter, beating
    analyst expectations. The Texas-based company said demand remained
    strong in China and Europe despite increased competition from
    traditional automakers entering the EV market."""",
    """NASA's Perseverance rover discovered new evidence of ancient
    water activity on Mars, according to a paper published this week.
    Scientists at the Jet Propulsion Laboratory said the findings
    strengthen the case that the planet once had a wetter climate."""",
    """Manchester United announced the signing of a new midfielder
    from Barcelona for a reported fee of 80 million euros. The 24-year-old
    player will join the club ahead of the new season starting in
    August."""",
]

```

```

import spacy
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
nlp = spacy.load('en_core_web_sm')
summarizer_tokenizer = AutoTokenizer.from_pretrained("sshleifer/distilbart-cnn-12-6")
summarizer_model = AutoModelForSeq2SeqLM.from_pretrained("sshleifer/distilbart-cnn-12-6")

```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable token authentication.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub.
config.json: 100% 1.80k/1.80k [00:00<00:00, 59.8kB/s]
tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 533B/s]
vocab.json: 100% 899k/899k [00:00<00:00, 19.4MB/s]
merges.txt: 100% 456k/456k [00:00<00:00, 12.9MB/s]
pytorch_model.bin: 100% 1.22G/1.22G [00:10<00:00, 149MB/s]
[transformers] Please make sure the generation config includes `forced_bos_token_id=0`.
Loading weights: 100% 358/358 [00:00<00:00, 6709.39it/s]
model.safetensors: 100% 1.22G/1.22G [00:16<00:00, 63.7MB/s]

```

```

def digest_article(text):
    inputs = summarizer_tokenizer(text, return_tensors="pt", truncation=True, max_length=1024)
    summary_ids = summarizer_model.generate(inputs["input_ids"], num_beams=4, max_length=40, min_length=10)
    summary_text = summarizer_tokenizer.decode(summary_ids[0], skip_special_tokens=True)
    doc = nlp(text)
    entities = {}

    for ent in doc.ents:
        entities.setdefault(ent.label_, set()).add(ent.text)

```

```
    return summary_text, entities
for i, article in enumerate(articles, 1):
    summary, entities = digest_article(article)

    print(f"--- Article {i} ---")
    print("Summary :", summary)
    print("Entities:")

    for label, names in entities.items():
        print(f"  {label}: {' '.join(names)}")

    print()
```

--- Article 1 ---

Summary : CEO Tim Cook said the device features a faster chip and improved camera . The phone will

Entities:

ORG: Apple

GPE: Cupertino

DATE: next month, Tuesday

PERSON: Tim Cook

MONEY: 999

--- Article 2 ---

Summary : The World Health Organization announced new guidelines aimed at reducing sugar consumption

Entities:

ORG: The World Health Organization

DATE: the next

decade, Monday

GPE: Geneva

CARDINAL: millions

--- Article 3 ---

Summary : Tesla reported record deliveries for the third quarter, beating analyst expectations .

Entities:

DATE: the third quarter

GPE: Texas, China

LOC: Europe

ORG: EV

--- Article 4 ---

Summary : NASA's Perseverance rover discovered new evidence of ancient water activity on Mars .

Entities:

ORG: NASA, the Jet Propulsion Laboratory

LOC: Mars

DATE: this week

--- Article 5 ---

Summary : Manchester United announce signing of Barcelona midfielder from Barcelona . The 24-year-old

Entities:

PERSON: Manchester United

GPE: Barcelona

CARDINAL: 80 million

DATE: 24-year-old, August

